

SQL Fundamentals - 16 June 2023

• Sorting statements

↳ ORDER BY statement

- We can sort our data either in Ascending Order (ASC) or Descending Order (DESC)

- Sorting does not remove or add anything new into our data - it just re-arrange the rows in our data.

(ASC)

• Ascending order - from smallest to biggest; lowest to highest; it's going up

(DESC)

• Descending order - its going down; from biggest to smallest; highest to lowest

Syntax:
SELECT *
FROM table_name
WHERE condition
ORDER BY column_name ASC;

Ascending order

Syntax:
SELECT *
FROM table_name
WHERE condition
ORDER BY column_name DESC;

Descending order

Students

Student_id	name	course	age	city
1	Thabo	DA	22	UHB
2	Sarah	DS	19	PTA
3	John	DA	23	DBN
4	Lerato	DA	20	JHB
5	Musa	DS	23	UHB

DA = Data Analytics

DS = Data Science

The answer is the name & age
but sorted according to age in ascending order

SELECT name,
age
FROM students
ORDER BY age ASC;

SELECT name,
age
FROM students
ORDER BY age DESC;

The answer is the name & age
but sorted according to age in descending order

• Aggregate Functions

↳ Returns a single value

↳ An aggregate function is used to perform a calculation on rows of a specific column and returns a single value / a single value per group

Function	Description
① Count ()	Counts the number of rows
② Sum ()	Calculates the total of a numeric column
③ Avg ()	Calculates the average of a numeric column
④ Max ()	Find the maximum value
⑤ Min ()	Find the minimum value

Employees

Empl_id	Name	Salary
101	John	20
102	Mary	40
103	David	35

```
- SELECT MIN (salary) AS min_salary  
  FROM employees; -> 

|            |
|------------|
| min_salary |
| 20         |

  
- SELECT MAX (salary) AS max_salary  
  FROM employees; -> 

|            |
|------------|
| max_salary |
| 40         |


```

alias
* AS is a way to rename a column

- we use it to rename